Project Title	**Stock Market**
Tools	Power BI, MS Excel, DAX
Technologies	Business Analyst
Project Difficulties Level	Intermediate

Dataset: The dataset is available in the given link.

Dataset Link

Project Overview

To analyze historical price trends and trading volumes for major tech stocks to identify market volatility and investment opportunities.

About Dataset Content

The dataset contains daily historical stock prices for four major tickers (AAPL, MSFT, NFLX, GOOG).

Key Columns in the Dataset:

Ticker: The stock symbol (AAPL, MSFT, NFLX, GOOG).

Date: The specific trading day (formatted as YYYY-MM-DD).

Open / Close: The price at the start and end of the trading session.

High / Low: The maximum and minimum prices reached during the day.

Adj Close: The closing price adjusted for dividends and splits.

Volume: The total number of shares traded.

Power BI Code

1. Top Layer

KPI Cards

TOTAL TRADING VOLUME

Value: 7.96B (Formatted as Billions).

DAX: Total Volume = SUM('stocks'[Volume])

AVERAGE CLOSE PRICE

Value: $215.38

DAX: Avg Close = AVERAGE('stocks'[Close])

MAX STOCK HIGH

Value: $373.83

DAX: Highest Price = MAX('stocks'[High])

2. Middle Layer

Key Charts

Price Trend by Date (Line Chart)

- **Logic:** This visual tracks the daily closing price for each stock over the 3-month period to identify market trends, growth patterns, and price dips.
- **DAX / Field:** Use the **Close** column.
- **Visual Setup:**
    - Visual Type: Line Chart.
    - X-Axis: Drag the Date column here (ensure it is set to Date, not Hierarchy, for a continuous line).
    - Y-Axis: Drag the Close column.
    - Legend: Drag the Ticker column. This creates a separate colored line for AAPL, GOOG, MSFT, and NFLX.

Daily Price Spread (Clustered Bar Chart)

- **Logic:** A direct comparison of the volatility (the gap between the highest and lowest price reached) across the 4 tickers.
- **DAX / Field:** Use the High and Low columns.
- **Visual Setup:**
    - Visual Type: Clustered bar chart.
    - Y-Axis: Drag the Ticker column.
    - X-Axis: Drag both the High and Low columns into this bucket.
    - Formatting: In the "Format" pane, set different colors for High

3. Bottom Layer

<u>Slicers</u>

Ticker Slicer

- **Logic:** Enables stakeholders to filter the entire dashboard (KPIs and Charts) for one or multiple specific companies (AAPL, MSFT, NFLX, or GOOG).
- **Field:** Use the Ticker column from the stocks.csv dataset.
- **Visual Type:** Slicer visual (Funnel icon).
- **Visual Setup:** Drag the Ticker field into the Field bucket.

Date Slicer

- **Logic:** Provides temporal control, allowing users to zoom into specific weeks or months within the February–May 2023 timeframe for trend analysis.
- **Field:** Use the Date column.
- **Visual Type:** Slicer visual.
- **Visual Setup:** Drag the Date field into the Field bucket.

Dashboard Creation

Load the Dataset

- Open Power BI Desktop.
- Click Get Data > Text/CSV.
- Select your stocks.csv file and click Open.
- In the preview window, click Load.

Create the KPI Cards (Top Row)

- **Total Volume:**
    - Select Visual: Click the Card icon in the Visualizations pane.
    - Add Field: Drag the Volume column into the Fields bucket.
    - Display: Ensure it displays 7.96B.
- **Average Close:**

    - Select Visual: Click the Card icon.
    - Add Field: Drag the Close column and change the aggregation to Average.
    - Display: Ensure it displays $\$215.38$.
- **Highest Price:**
    - Select Visual: Click the Card icon.
    - Add Field: Drag the High column and change the aggregation to Maximum.
    - Display: Ensure it displays $\$373.83$.

Create the "Price Trend by Date"

- Select Visual: Click the Line Chart icon.
- X-Axis: Drag the Date column here.
- Y-Axis: Drag the Close column here.
- Legend: Drag the Ticker column here to show individual lines for AAPL, GOOG, MSFT, and NFLX.

Create the "Volume by Ticker"

- Select Visual: Click the Clustered bar chart icon (horizontal bars).
- Y-Axis: Drag the Ticker column here.
- X-Axis: Drag the Volume column here.
- Sort: Click the three dots (...) on the chart and select Sort axis > Volume to ensure the highest trading activity is at the top.

Create the "Daily Price Spread"

- Select Visual: Click the Clustered bar chart icon.
- Y-Axis: Drag the Ticker column here.
- X-Axis: Drag both the High and Low columns into this bucket.

Add Interactive Filters (Slicers)

- Select Visual: Click the Slicer icon (funnel/filter).
- Field: Drag the Ticker column into it.
- Repeat: Create another Slicer and drag the Date column into it.
- Settings: In the "Format" pane, change the slicer style to Dropdown to save space and keep the dashboard clean.

Save & Share

- Go to File > Save.

Conclusion:

This dashboard successfully transforms raw historical stock data into actionable market insights, identifying the major price trends and trading volumes across leading tech companies. By centralizing these financial KPIs and implementing interactive filtering, the report empowers stakeholders to perform comparative analysis and optimize investment strategies through data-driven decision-making.

Output

STOCK MARKET
MARKET PRICE
275.04
TOTAL VOLUME
2bn
DAY HIGH
17.22K
DAY LOW
16.87K
Market Performance Trends
400
300
200
100
0
Price
Mar 2023
Apr 2023
May 2023
Date
Average of Volume by Date and Ticker
Ticker MSFT
Average of Volume
50M
0M
Mar 2023
Apr 2023
May 2023
Date
NETFLIX
AAPL
GOOG
MSFT
NFLX